

Perishable Lot-Sizing with Period-Dependent Shelf Lives

A Compact Mixed-Integer Programming Model

Indexing and Sets

- Periods: $T := \{1, 2, \dots, \mathbf{T}\}$.
- Items: I . Typical indices: $i \in I$, $t, u \in T$ with $t \leq u$.

Arc Construction (Perishability)

Age constraint. If a unit of item i is produced in period t and consumed in period $u \geq t$, its *age* at consumption is $\text{age}(t, u) := u - t$. With a period-dependent shelf life $m_{i,t} \in \mathbb{Z}_{\geq 1}$, the unit is usable only while

$$\text{age}(t, u) \leq m_{i,t} - 1 \iff u \leq t + m_{i,t} - 1.$$

Reachable consumption periods from t . For each (i, t) define

$$\Gamma_{i,t} := \{u \in T \mid t \leq u \leq \min\{\mathbf{T}, t + m_{i,t} - 1\}\}.$$

These are exactly the periods in which production at t may still be consumed.

Feasible origins for a demand period u . Dually, for each (i, u) define

$$\Omega_{i,u} := \{t \in \{1, \dots, u\} \mid u \leq t + m_{i,t} - 1\}.$$

These are exactly the production periods that can supply demand at u .

Arc set. The model contains a decision variable only for pairs (t, u) that satisfy perishability:

$$A := \{(i, t, u) \in I \times T \times T \mid u \in \Gamma_{i,t}\} = \bigcup_{i \in I} \bigcup_{t \in T} \{i\} \times \{t\} \times \Gamma_{i,t}.$$

Equivalently, $(i, t, u) \in A$ iff $t \in \Omega_{i,u}$.

Edge cases.

- If $m_{i,t} = 1$, then $\Gamma_{i,t} = \{t\}$: production at t must be consumed immediately.
- If $m_{i,t} \geq \mathbf{T} - t + 1$, then $\Gamma_{i,t} = \{t, t + 1, \dots, \mathbf{T}\}$: no effective expiration within the horizon.

Tiny example (one item, $T=7$). Suppose $(m_{i,3}, m_{i,4}, m_{i,5}) = (2, 4, 1)$. Then:

$$\Gamma_{i,3} = \{3, 4\}, \quad \Gamma_{i,4} = \{4, 5, 6, 7\}, \quad \Gamma_{i,5} = \{5\}.$$

Thus, arcs $(i, 3, 5)$ or $(i, 5, 6)$ do *not* exist in A .

Parameters (all nonnegative unless noted)

$d_{i,u}$	demand of item i in period u .
c_i^{var}	variable production cost for item i .
h_i	unit holding cost per period for item i (applied to age $u-t$).
s_i	setup cost for item i at a production period.
b_i^{var}	backlog penalty cost for item i (used only if backorders are allowed).
cap_t	production capacity available in period t .
$m_{i,t} \in \mathbb{Z}_{\geq 1}$	period-dependent shelf life (max allowed age) for production of i in t .
$M_i := \sum_{u \in T} d_{i,u}$	(tight big- M for setup linking).

Decision Variables

$x_{i,t,u} \in \mathbb{Z}_{\geq 0}$	quantity of item i produced at t and consumed at u , for $(i, t, u) \in A$.
$y_{i,t} \in \{0, 1\}$	setup indicator for item i in production period t .
$b_{i,u} \in \mathbb{Z}_{\geq 0}$	backlog at the end of period u for item i (only if backorders are used).

Objective

$$\begin{aligned} \min \quad & \sum_{(i,t,u) \in A} (c_i^{\text{var}} + h_i (u-t)) x_{i,t,u} + \sum_{i \in I} \sum_{t \in T} s_i y_{i,t} \\ & + \text{(include only if backorders are used)} \sum_{i \in I} \sum_{u \in T} b_i^{\text{var}} b_{i,u}. \end{aligned} \quad (1)$$

Constraints

$$\text{Capacity:} \quad \sum_{\substack{(i,t,u) \in A \\ t=\tau}} x_{i,t,u} \leq \text{cap}_\tau \quad \forall \tau \in T, \quad (2a)$$

$$\text{Setup linking (tight } M_i\text{):} \quad \sum_{u \in \Gamma_{i,t}} x_{i,t,u} \leq M_i y_{i,t} \quad \forall i \in I, \forall t \in T. \quad (2b)$$

Demand balance — choose the block matching your setting. A) *No backorders:*

$$\sum_{t \in \Omega_{i,u}} x_{i,t,u} = d_{i,u} \quad \forall i \in I, \forall u \in T. \quad (2c)$$

B) *With backorders:*

$$\sum_{t \in \Omega_{i,1}} x_{i,t,1} + b_{i,1} = d_{i,1} \quad \forall i \in I, \quad (2d)$$

$$\sum_{t \in \Omega_{i,u}} x_{i,t,u} + b_{i,u} = d_{i,u} + b_{i,u-1} \quad \forall i \in I, \forall u \in \{2, \dots, \mathbf{T}\}. \quad (2e)$$

Domains (integrality): $x_{i,t,u} \in \mathbb{Z}_{\geq 0}$ $((i, t, u) \in A)$, $y_{i,t} \in \{0, 1\}$, and if used, $b_{i,u} \in \mathbb{Z}_{\geq 0}$.

Remarks

- Arcs (i, t, u) exist *only* when $u \in \Gamma_{i,t}$. This is how perishability is enforced without additional constraints.
- The demand and setup constraints naturally use the companion sets: sums over $u \in \Gamma_{i,t}$ and over $t \in \Omega_{i,u}$.
- The choice $M_i = \sum_u d_{i,u}$ is a standard tight bound for linking x to y .